

Glossa Lab: A Computational Toolkit for Ancient Script Analysis and Decipherment

Project Report and Collaboration Proposal

BitConcepts — 2026-04-01

1. Project Overview

Glossa Lab is a proprietary, cross-platform research toolkit for computational analysis and decipherment of ancient scripts, developed by BitConcepts. Built in Python with a FastAPI backend, it combines statistical analysis, structural pattern detection, and automated decipherment in a single integrated platform. The toolkit incorporates patented technologies and is not publicly available at this time.

The toolkit was designed with the Indus script as its primary target, but has been validated on synthetic ciphers and the historically deciphered Ugaritic script to establish baseline performance before attempting the undeciphered Indus corpus.

2. Analysis Pipelines (11 total)

Pipeline	Function	Status
block_entropy	Rao et al. (2009) normalised block entropy H_N	Implemented
char_freq	Zipf distribution, rank-frequency, Zipf exponent	Implemented
kandles	Phonetic color-coding and grids (Merkur patent)	Implemented
positional	Position-specific sign frequencies (initial/medial/terminal)	Implemented
sign_cluster	Distributional clustering (Kober method)	Implemented
paradigm	Inflectional paradigm detection (Ventris/Kober)	Implemented
cooccurrence	Sign co-occurrence network + community detection	Implemented
numerals	Numeral sign identification from distributional behavior	Implemented
decipher	Substitution cipher cracking (hill climbing + trigrams)	Implemented
hypothesis	Iterative hypothesis-driven decipherment engine	Implemented
CPSC projection	Multi-constraint decipherment (optional module)	Implemented

Table 1. Analysis pipelines.

3. Validated Results

3.1 Ugaritic Decipherment — Real Ancient Script (96.7%)

Our primary validation was on **real Ugaritic text** — the Baal Cycle (KTU 1.1–1.6), a cuneiform alphabetic script from c. 1400–1190 BCE that was deciphered in the 1930s. We encoded 83 lines of genuine Ugaritic inscriptions with opaque sign IDs (U01–U30) to simulate the undeciphered state, then ran our decipherment engine against a language model built from the known transliteration.

Result: 29 of 30 signs correctly identified (96.7% accuracy). The single miss (sign U30/s2) is the rarest sign in the Ugaritic alphabet, appearing zero times in our corpus. Top-5 most frequent signs: 5/5 correct. Kandles phonetic confidence: 1.000.

3.2 Synthetic Cipher Benchmark (100%)

As a controlled baseline, we created a toy language (21 phonemes, CVC words, 3 noun cases, 2 verb tenses, SOV word order) and encrypted it with a random substitution cipher. The engine recovered all 21 phoneme mappings: **100% accuracy**.

Test	Data Type	Accuracy	Details
Ugaritic Baal Cycle	REAL ancient text	29/30 = 96.7%	83 lines, 30 cuneiform signs, c. 1400 BCE
Synthetic cipher	Synthetic benchmark	21/21 = 100%	21 phonemes, 500 inscriptions

Table 2. Decipherment accuracy on real and synthetic data.

3.3 Block Entropy — Real Multi-Language Corpora

We replicated the block entropy analysis from Rao et al. (2009) across 9 corpora using **real text** for English (Melville), Tamil (Thirukkural), Sanskrit (Rigveda), and DNA (human beta-globin gene), plus synthetic baselines. Results confirm the published entropy ordering: linguistic systems cluster together, distinct from non-linguistic systems.

3.4 Indus Script Hypothesis Test (Synthetic Corpus)

We tested proto-Dravidian vs Vedic Sanskrit as competing target language hypotheses on a **synthetic** Indus corpus (6,823 signs, 417 unique) that reproduces published statistical properties (Yadav et al. 2010). This is NOT the actual M77/ICIT data — validation on real inscriptions is the critical next step.

Hypothesis	Score	Word Matches	Kandles
Proto-Dravidian	297.0	28 (28% coverage)	0.985
Vedic Sanskrit	77.0	6 (4.9% coverage)	0.976

Table 3. Hypothesis engine results (synthetic Indus corpus). Proto-Dravidian scores 4× higher. Pending validation on real data.

4. What We Need: The ICIT Corpus

Our toolkit is validated and ready. The single critical blocker is access to the actual Indus corpus in machine-readable form. The ICIT database (Wells & Fuls) contains 4,537 inscribed objects with 5,509 texts and 19,616 sign occurrences — exactly the data our pipelines are designed to process.

What we would do with ICIT access:

1. Run all 11 analysis pipelines on the complete corpus
2. Validate our synthetic results against real sign distributions
3. Run the hypothesis engine (Dravidian vs Sanskrit) on real data
4. Apply numeral identification to reduce the phonetic search space
5. Run positional analysis to classify all 676 signs by function
6. Detect paradigms (inflectional patterns) across real inscriptions
7. Build co-occurrence networks to identify sign communities
8. Generate comprehensive PDF reports with all findings

We have already built a Fuls-notation parser that handles the +sign-sign-sign+ format used in the published corpus volumes, including metadata extraction (findspot, object type, iconography, reading direction).

5. Decipherment Methodology

Stage 1 — Structural analysis: Block entropy confirms the script is linguistic. Positional analysis classifies signs by grammatical function. Paradigm detection reveals inflectional patterns. Numeral identification reduces the search space.

Stage 2 — Hypothesis testing: Competing target language models (proto-Dravidian from DEDR roots + Old Tamil; Vedic Sanskrit from Rigveda) are tested against the corpus. The hypothesis that produces more word matches and better paradigm regularity is promoted.

Stage 3 — Iterative decipherment: Confident sign mappings are locked. Remaining signs are optimised against the target language model using bigram/trigram hill climbing with Kandles phonetic validation. Each iteration reduces the unknown space.

6. Technical Details

102 automated tests • Python 3.12 • FastAPI backend • SQLite database • CI on Windows/Linux/macOS • PDF report generation • Cross-platform • Proprietary — incorporates patented technologies (US 2024/0248922 A1, US Provisional 63/980,251)

References

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